THE MALAWI NATIONAL EXAMINATIONS BOARD

2005 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

AGRICULTURE

Subject Number: M012/II

Time Allowed: 2 hour sessions
8:30 am onwards

Thursday, 13 October

PAPER II

(100 marks)

Practical

Instructions

1. This paper contains 4 pages. Please check.

2. Answer **all** questions using the Answer Book provided. Each question carries 20 marks. The marks are indicated against each section of the question.

3. Write your **Examination Number** on **all** pages of your Answer Book.

5. At the end of the examination, hand in your Answer Book to the invigilator.

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2005

1. You are provided with two specimens labelled **A** and **B**. Use them to answer questions that follow.

   a. Identify the specimens labelled **A** and **B**. (2 marks)

   b. Draw specimen **B** and label any **three** parts. (5 marks)

   c. (i) Which of the two specimens is used as an animal feed? (1 mark)

   (ii) Explain any **two** ways of improving the feeding value of the specimen named in **c(i)**. (4 marks)

   (iii) Which class of animals is best suited to feeding on the specimen named in **c(i)**? (1 mark)

   d. (i) Which part of specimen **B** is used for grafting? (1 mark)

   (ii) Describe how the part named in **d(i)** can be grafted. (2 marks)

   e. Explain any **two** functions of a leaf to a growing crop. (4 marks)

2. You are provided with Specimens **M** and **N**. Use them to answer questions that follow.

   a. Identify the Specimens **M** and **N**. (2 marks)

   b. State any **two** processing activities through which Specimens **M** and **N** undergo in readiness for storage. (4 marks)

   c. Explain any **two** conditions which are required in the storage of Specimen **M**. (4 marks)

   d. Explain any **two** ways in which Specimen **N** is important in terms of food security. (4 marks)

   e. (i) State the nutritive value of Specimen **N**. (2 marks)

   (ii) Give any **two** products obtained from processing Specimen **N**. (2 marks)

   f. Describe **one** way in which processing companies that use Specimen **M** help to ensure food security. (2 marks)

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3. **Table 1** shows information on feeding of layers. Use it to answer questions that follow.

Table 1

AVAILABLE FEEDS	PROTEIN CONTENT
Maize meal	7%
Soya bean mean	40%

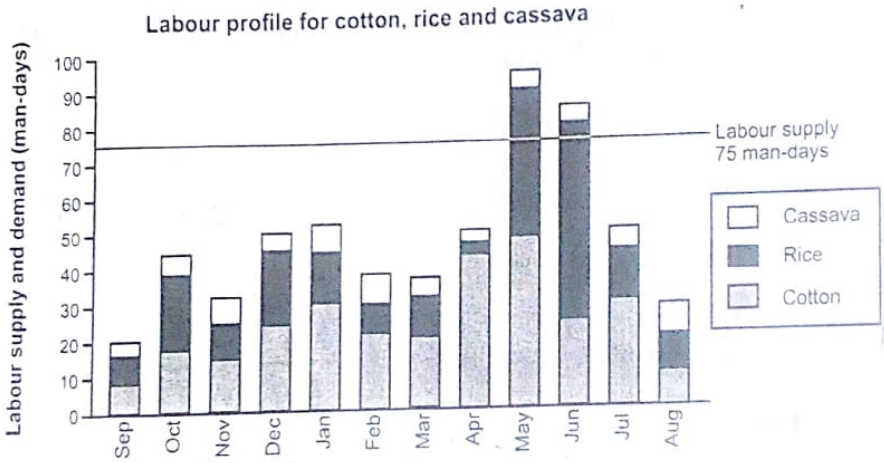
a. Using a pearson square method, formulate a 990 kg ration for layers containing 21% protein. Show your calculation. **(8 marks)**

b. Explain any **three** factors which should be considered when balancing a ration. **(6 marks)**

c. Describe any **two** functions of a soya bean meal to layers. **(4 marks)**

d. Suggest any **two** types of nutrients which could be added to the feed in **(a)** to improve quality. **(2 marks)**

4. The **figure below** is a bar graph showing labour profile for cotton, rice and cassava. Use it to answer questions that follow.



a. (i) During which months was labour insufficient? **(2 marks)**

(ii) In which month was labour demand the lowest on the farm? **(1 mark)**

b. (i) Calculate the total amount of labour needed to meet demand in the months mentioned in **a(i)**. **(3 marks)**

(ii) Explain any **two** ways in which a farmer could solve the problem in **b(i)**. **(4 marks)**

c. Name **one** crop which had the highest labour demand and another which had the least labour demand during the year. **(2 marks)**

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4. (Continued)

d. Explain any **two** ways in which knowledge of labour demand for different enterprises is important. **(4 marks)**

e. (i) Give **one** point to explain what would happen to labour demand if all the land on which cotton was grown was used for growing rice. **(2 marks)**

(ii) Other than labour, state any **two** farming resources which are used in production. **(2 marks)**

5. **Table 2** shows growth rate of bean seedlings. Use it to answer questions that follow.

Table 2

	DAYS	1	2	3	4	5	6	7	8
Growth rate (mm)	Seedlings from plump seeds	17	25	30	36	50	64	82	96
	Seedlings from shrunken seeds	14	16	22	25	30	33	49	56

a. Draw a line graph with **two** curves; **one** on seedlings which germinated from plump seeds and **another** on those from shrunken seeds. **(9 marks)**

From the graph

b. (i) Give the mean shoot length of each set of seedlings after $5\frac{1}{2}$ days of germination. **(2 marks)**

(ii) Describe the growth rate of seedlings from plump and shrunken seeds within the first day of germination. **(2 marks)**

(iii) Explain why there is a difference in the curves on the graph after **day 4**. **(2 marks)**

c. State any **three** conditions which should have been held constant during the experiment. **(3 marks)**

d. State **one** conclusion you would make from the results to show why it is necessary to select seeds before planting. **(2 marks)**

END OF QUESTION PAPER

NB: This paper contains 4 pages.

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